ECDLP lead atlas: first full review (2026-10-24)

Question. Which ECDLP leads have the four repositories (crypto-autoresearcher, crypto, cryptanalysis, ecc-challenges) already examined? What decided each one, and what is left? The aim is a state in which a fresh clone can tell, by computation, that everything has already been examined.

Scope and method. Seven parallel surveys, one per lead family (generic rho, binary IC, the IC solving engine, prime-field IC, transfers, curve structure, and exotic or statistical ideas). They read the summaries in all four repositories: decision and evidence records, findings, problem records, study READMEs and conclusions, leaderboards and benchmark indexes. Each survey drafted lead rows and mapped every ECDLP-area ledger record to a lead. I merged the rows and resolved cross-family duplicates. The original survey ran `tools/lead_atlas.py --check --strict-paths` in its checkouts. A later check against published crypto main found six study citations that exist only on open crypto PR #2495 and two walk citations that exist only on open ecc-challenges PR #56, plus one symlink alias that is corrected in the lead row. The eight citations are now pinned to their source commits; the [provenance problem](../../../problems/ecdlp-lead-atlas-unmerged-evidence.md) tracks their replay and four-repository verification. The original survey also folded in cryptanalysis `main` work that its first pass could not see (its cloned default branch is a stale lineage; see Limitations).

Status of each statement. Dispositions summarize the cited records; eight citations still point to unmerged source PRs and are provisional. This report makes no new measurement and changes no hypothesis, experiment or goal status. Numbers are copied from the cited records, not independently replayed here. Rankings are this review's *proposal*.

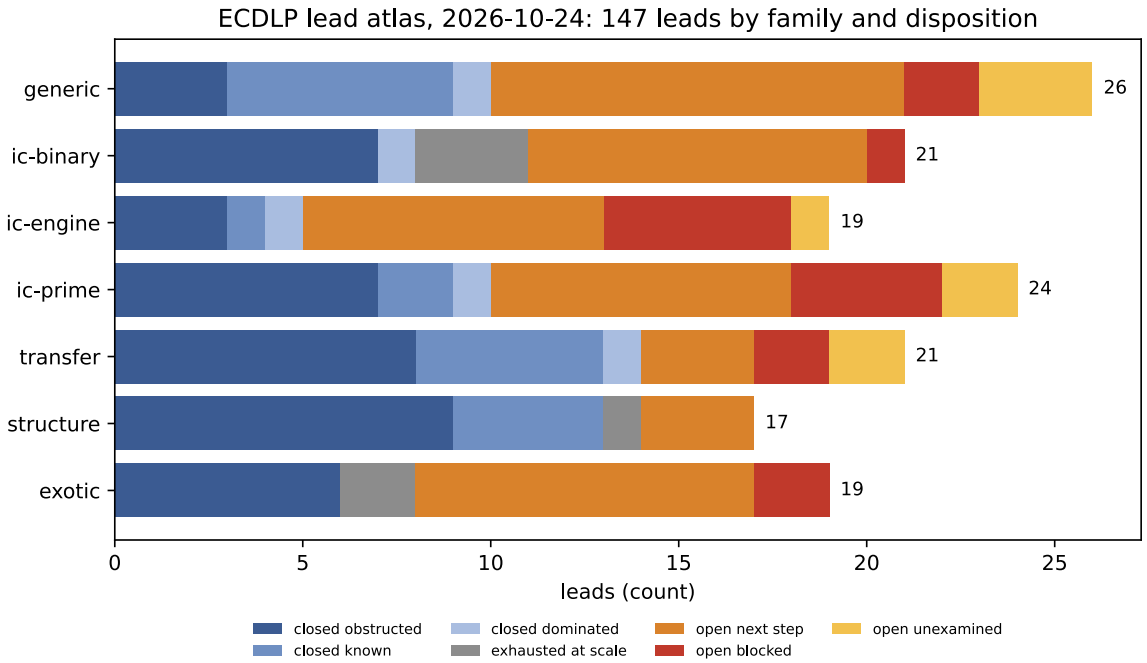


Figure 1: Leads by family and disposition

The chart counts filed dispositions. It does not certify their cited experiments.

Result

- **147 leads**: 72 closed, each with a named obstruction (43 obstructed, 18 known, 5 dominated, 6 exhausted at scale), 51 open with a decided next step, 16 blocked and 8 unexamined.
- **Coverage 2049/2049**: every hypothesis and idea in an ECDLP area of the ledger maps to a lead (`knowledge/leads/ecdlp/coverage.tsv`). 264 rows are mapped `NONE` because they are not about the discrete log: rank records, code-based and PQC strays, and arithmetic trivia.
- **Review pinned**: `REVIEW_STATE.json` records the original review's default-branch commits: autoresearcher `4fa15469d`, crypto `da3a1b5d5`, cryptanalysis `9760a763d`, ecc-challenges `3a71596ba`. Its 50/9 tally is historical. The committed rows now tally 51/8 after the hybrid guess-and-determine lead advanced. At this regeneration, `--delta` reports changed watched paths in autoresearcher; the three sibling repositories were not checked out here, so their later changes remain unverified.

family	leads	closed	open, next step	blocked	unexamined
generic (rho, kangaroo, BSGS, pool)	26	10	11	2	3
ic-binary	21	11	9	1	0

ic-engine	19	5	8	5	1
ic-prime	24	10	8	4	2
transfer	21	14	3	2	2
structure	17	14	3	0	0
exotic	19	8	9	2	0

The structure and transfer families are nearly exhausted. The theorems that close them are Tate's isogeny theorem, the Balasubramanian–Koblitz embedding-degree condition, the genus-g index-calculus floor `q^{2-2/g} ≥ q`, and the triviality of homomorphisms from a prime-order group into an ℓ-group (ℓ ≠ the group order). The open frontier is index-calculus engineering and the exponent questions inside the solving engine.

Distance remaining to matched rho (measured, copied)

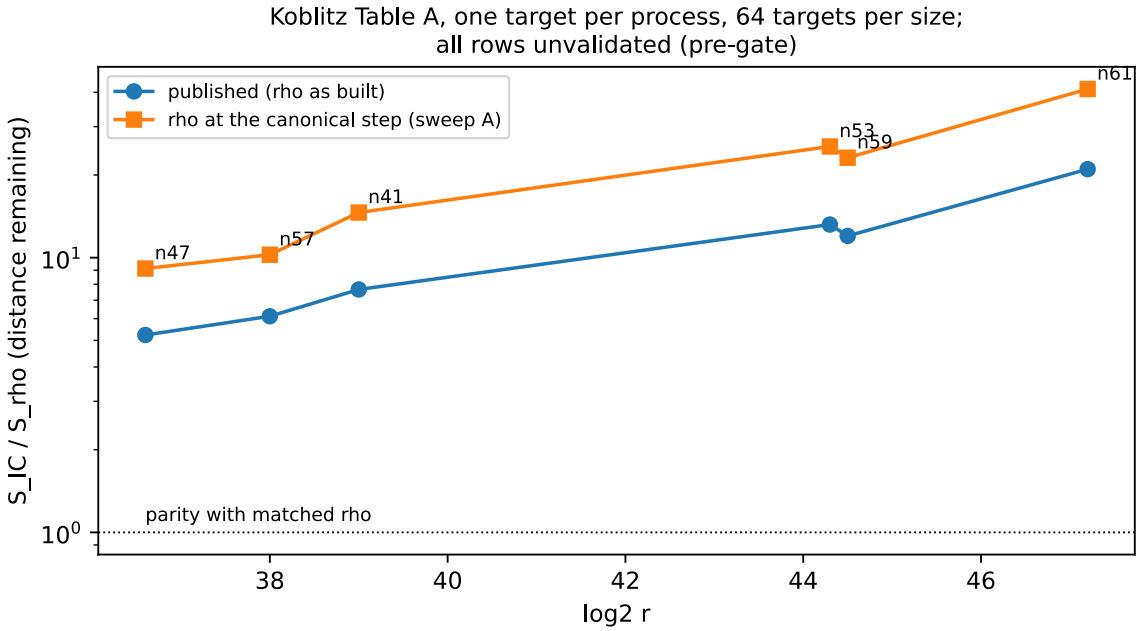


Figure 2: Table A gap, published and at the canonical rho step

board	best row	IC / matched rho	caveat
crypto Table A (Koblitz, one target)	n47	5.23x published, 9.13x at the canonical step	rho priced "as built"; sweep A re-prices at 9.13–41.1x
crypto Table B (prime field)	fp12	5.47x	gap widens with log2 r to 13.5x at fp24
crypto Table C (Koblitz, 32 targets)	n41	3.62x	rho from a different session
cryptanalysis Koblitz ratchet	n83 rank-finder row	3.92x cold vs tuned rho	wall time on one host class, factor-base construction uncharged, §5a band unresolved
generic rho vs floor	Koblitz signed-Frobenius strong rho	1.089x floor	canonicalisation counted, not charged

Every IC row is still "unvalidated (pre-gate)". No row is below its reference. The online "IC beats rho" readings (n61 13.9x, N83 18.5x and similar) compared IC against rho with an empty distinguished-point table. They are not results: at equal advice, Bernstein–Lange rho is 53–317x faster online.

Next steps, ranked across families (proposal)

1. **Port the cryptanalysis n=83 engineering ratchet to crypto Table A, then re-run Table A** with tuned K and production free relations (`LEAD-ICB-koblitz-setup-phase-cost`). Set-up is 98.5–99.7% of Table A's S. Off-board figures (K retune 0.46x at n41, free relations 0.960x) and the ten stacked n=83 producers indicate that this is the cheapest verified IC-baseline improvement. It is constant-level only.

2. **Validation gate for the IC boards.** No row can land until it carries a replay certificate, a validator report and the §5a band (crypto CLAUDE.md, gate of 2026-10-19). This blocks every other improvement from counting.
3. **Descent closure degree at (m, l) = (3, 7) and (4, 7)** (`LEAD-ICE-descent-closure-degree-at-scale`, blocked on a host with at least 250 GB). This is the only exponent-level question left for prime-degree binary curves: degree 4 puts the ECC2K-130 m=4 row about +7.6 bits over rho, degree 5 about +10.3 bits, and degree 6 or more closes it. The new `LEAD-ICE-hybrid-guess-determine-l7` probes the same question without that host.
4. **Class quotient in the ecc-challenges v2 worker** (`LEAD-RHO-pool-class-quotient`). Negation (√2) and, on Koblitz campaigns, signed Frobenius (√(2m)) are missing from all 23 non-ECC2K-130 campaigns. The win is certain engineering.
5. **Price the native-XOR SAT arm** (`LEAD-ICE-sat-xor-pdp-oracle`). Its S/rho of 0.854 at n=17 Koblitz holds only because solver conflicts are unpriced. One calibration decides the lead either way.
6. **Optimal Coppersmith shift set** (`LEAD-ICP-coppersmith-smallroot-semaev`). The recorded closure "by 0.048" used one Jochemsz–May strategy, with 0.159 of slack unused. Ryan's (EUROCRYPT 2025) optimal shifts at (m, w) = (4, 2) are a calculator run with a decisive outcome.
7. **Joux–Vitse sieve on the GF(p^6) weak class, with full accounting** (`LEAD-TRF-jv-sieve-fp6-weak-class`). This is the only transfer ever measured below strong rho (0.011x including the walk), but its comparisons were withheld because the tolerances and hash costs were not frozen. It reproduces a 2012 result and applies to the weak class only.
8. **Exotic lanes with a decisive cheap test**: the bucket variance W of the four-list O2 kernel (`LEAD-EXO-four-list-ktree-join`), and claims K1 and K2 of the Miller cocycle (`LEAD-EXO-miller-pairing-cocycle`).

New leads added by the original review: `LEAD-ICE-linearization-excess-fb-prior`, `LEAD-ICE-hybrid-guess-determine-l7` and `LEAD-TRF-teske-trapdoor-fp6-weak-class`. The hybrid lead now has measured evidence and an `open_next_step` disposition. The surveys proposed nine more candidates; `--match` found each one already present as a lead or a KR row. That is the gate doing its job.

Integrity findings for the Coordinator (observed; no status changed here)

- **Ledger backlog.** 971 of the 1,031 ECDLP-area hypotheses (94%) have no terminal status: 373 approved, 308 specified, 240 proposed, 47 analyzed, 3 other. The surveys counted more than 100 approved hypotheses that are arithmetic trivia or Stage-0 identity checks ("Eight times six is 48"). They inflate counts without touching a lead.
- **Stale or contradicted documents an agent would re-propose from:**
  - `crypto/reports/Index calculus leads next steps.md` calls ten items unexecuted that `research/ic_next10_20261014/` ran.
  - `analysis/o2-sum-compatible-filters/README.md` says the four-tree route is closed; KN-FIND-ffe1df says it is not.
  - Two monodromy findings conflict: KN-FIND-a1f3c2 against KN-FIND-bdcd60.
  - Two elliptic-net findings conflict: KN-FIND-002 against KN-FIND-b7e091.
- **Status disagrees with evidence:**
  - H-IT-001 was closed on Tate's theorem (EV-IT-511f3d, strong), then reset to `weakened` with no new evidence.
  - H-ENDO-001 is `approved` against strong contradicting evidence.
  - H-QSP-cd0c90 expired with no result but is still `approved`.
- **Results without evidence records:** EXP-GRUMPY-cd7b82 stage 1, RUN-AUXIN-fd1edc, five KIC runs and the CZLIFT round (20 validated trials).
- **Instruments that could not fail:**
  - The XEDN `max|coeff| = 1` statistic.
  - ECTD meters: identical on all 32 curves.
  - The volcano census's ±10% IC margin. It could not see the measured 5.8% ℓ=2 floor deficit.
- **Overbroad closure:** KR-RHO-755e35 and EV-IT-511f3d say that isogeny walks never find weak curves. That is true over prime fields and false over composite-degree binary fields, where it is exactly how the GHS walks and Teske's trapdoor work.
- **Mislabelled regression:** the leaderboard's "96 root-finder replay mismatches" are inherited-F4 `solver_ops` rows (BMI2 rewrite, F4 pricing changes), per validation sweep D. They are not root-finder rows.
- **Repository hygiene:** the cryptanalysis GitHub default branch is a stale lineage (`claude/cryptanalysis-repo-setup-p59kek`, about 530 commits not on `main`). Sessions cloning the default read old results. Four atlas evidence paths exist only there, so they are pinned with `ref: c273e621`.

How the harness uses this

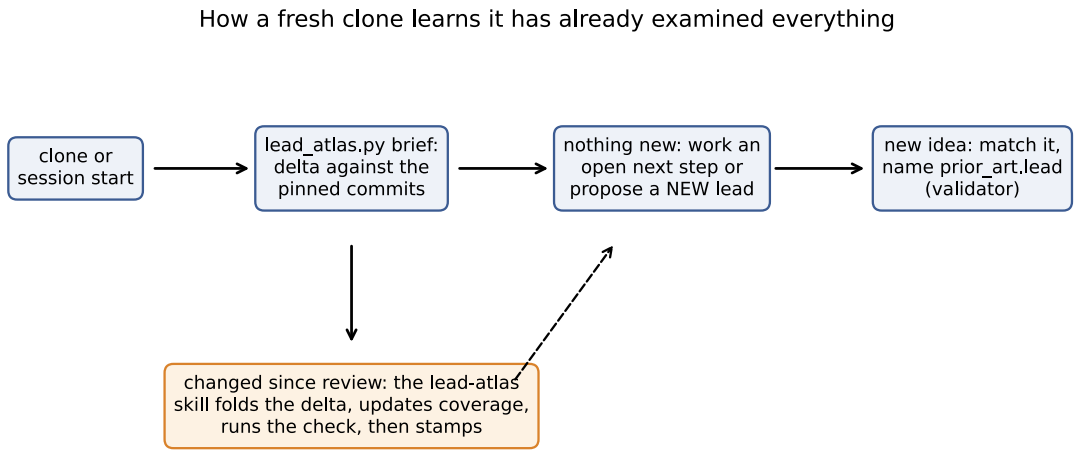


Figure 3: Harness loop

- **Session start.** All three research repositories print `tools/lead_atlas.py --brief`. When `--delta` is empty, the brief says that everything was already examined and points at the open next steps.
- **Ideation.** `/propose-ideas` pastes `--render` and runs `--match` on each returned idea. From IDEA-20261025 on, the validator requires `prior_art.lead`: the lead the idea advances, or `new` with a reason.
- **Review.** The `lead-atlas` skill folds a non-empty delta into leads, then runs `--check` and `--stamp`. CI and `make check-ledger` run `--check`.
- **Saturation.** No unexamined lead, no unmapped record, and nothing new since review. At this regeneration, only the mapping gate holds in this checkout: eight leads remain unexamined and `--delta` reports changed watched paths. Sibling-repository changes were not checked here. Saturation is reported, never used to pause a goal.

Limitations

- **Survey depth.** The surveys read summaries, not every raw run. A disposition is only as good as the decision or evidence record it cites. The coverage mapping is a scripted classification with hand overrides, and the boundary between `OTHER:ic-prime` and exotic among RQ-ECDLP-002 Stage-0 records is noisy.
- **Cryptanalysis lineage.** The first pass read cryptanalysis at its stale default branch. The `main`-only ECDLP material was folded in afterwards: the n=83 IC ratchet, the genus-2 cover and speed studies, and the n=83 surfaces run. Tooling-only changes were not.
- **Unvalidated sources.** Every gap figure is copied from unvalidated (pre-gate) rows.
- **Unmerged crypto evidence.** Six study citations are pinned to [PR #2495](https://github.com/aburan28/crypto/pull/2495) at `1397a09db5fb48edc2f9c2edca78035a2e4da5ca`, because those paths were absent from the atlas's pinned crypto main and current published crypto main. Their source paths exist at that PR head, but its current-head checks and an independent raw replay remain open. Two walk citations are pinned to open [ecc-challenges PR #56](https://github.com/aburan28/ecc-challenges/pull/56) at `1f49962624b13fbed4a3e9ad980d757f5fa5bc59`, which is currently conflicted. The four cryptanalysis branch-ref paths and the eight `ecc-challenges` paths were verified through the GitHub content API at their cited commits ([16-file path receipt](source-path-check.json)), but not through a complete local four-repository replay.

Sources: the lead rows (`knowledge/leads/ecdlp/<family>/`), each citing its own records; the snapshot of all rows (`atlas-snapshot.md`); the scoped source-path receipt (`source-path-check.json`); figures (`figures.py`, drawn only from committed rows and the cited leaderboard and sweep values).